

Exchange Identities and Computational Diagnostics for Triangle-Free Graph Bipartization

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Abstract

For a graph G , let

$$d(G) = |E(G)| - \text{MaxCut}(G)$$

be the minimum number of edges whose deletion makes G bipartite. A classical conjecture of Erdős asserts that every triangle-free graph on N vertices satisfies $d(G) \leq N^2/25$; for $N = 5k$ this is $d(G) \leq k^2$.

We study a five-vertex induction strategy whose target is

$$\exists X \subseteq V(G), |X| = 5 : \quad d(G) - d(G - X) \leq 2k - 1.$$

The paper develops an exact coordinated-flip identity, a corresponding selection inequality, elementary edge/star/biclique interaction corollaries, and reproducible finite certificates for reduced extension models. We also record exact computational diagnostics showing that stronger fixed-five-set statements can fail even when the existential induction target is easily repaired by changing the five-set.

A correction to an earlier draft is important. If

$$\ell(X) = 2d(G[X]) + e(X, G - X),$$

then the empty flip already proves the induction target whenever $\ell(X) \leq 4k - 1$. Thus at $k = 3$ the automatic range extends through $L(G) = 11$, where $L(G) = \min_{|X|=5} \ell(X)$. The previously reported $L = 11$ exchange audit therefore remains a valid structural computation, but is not needed to prove the induction inequality in that regime.

We place the project in the established language of frustration-critical signed graphs and weakly bipartite signed graphs. The full Erdős conjecture remains open; this manuscript is a structural and computational investigation of one proof architecture, not a proof of the conjecture.

1 Introduction

For a finite simple graph G , define

$$d(G) = \min\{|F| : F \subseteq E(G), G - F \text{ is bipartite}\} = |E(G)| - \text{MaxCut}(G).$$

Erdős conjectured that every triangle-free graph on N vertices satisfies

$$d(G) \leq \frac{N^2}{25}.$$

The balanced blow-up of C_5 shows that the constant $1/25$ would be sharp. Balogh, Clemen and Lidický proved the global bound $N^2/23.5$ and proved the sharp $N^2/25$ bound in two density ranges

[1]. A recent computer-assisted preprint of Ferudun proves the sharp value for every multiple of five up to $N = 200$ [2]. Consequently, the small finite cases appearing below are used as validation and structure-finding devices, not as new extremal evaluations.

The induction target studied here is the following.

Conjecture 1 (Five-set induction target). *For every triangle-free graph G on $5k$ vertices, there exists a five-set $X \subseteq V(G)$ such that*

$$d(G) - d(G - X) \leq 2k - 1.$$

If this statement holds for all k , induction from $k = 1$ gives

$$d(G) \leq 1 + 3 + \cdots + (2k - 1) = k^2.$$

Thus the problem becomes one of selecting a good five-set and extending an optimal coloring of the remaining graph.

This paper has two goals. First, we develop exact identities that separate boundary imbalance from internal recoloring gain. Second, we use finite certificates to diagnose which strengthenings of the induction are plausible and which are false.

2 Literature context and scope

The quantity $d(G)$ is the edge odd-cycle-transversal number, equivalently the MaxCut defect. In the language of signed graphs it is a special case of the frustration index. Cappello and Steffen study k -critical signed graphs, where deleting any edge lowers the frustration index [3]. We therefore do not claim the general notion of edge-criticality as new.

Likewise, the odd-circuit-cover polyhedron and its integrality theory are classical. Guenin's theorem characterizes weakly bipartite signed graphs by exclusion of an odd- K_5 minor; Schrijver gave a short proof, and Geelen–Guenin developed related packing consequences [4, 5]. These results are used here only as background for possible obstruction viewpoints.

The contribution claimed in this manuscript is narrower: the specific five-set induction framework, the coordinated-flip selection identity and its use as a diagnostic tool, exact reduced-state certificates, and reproducible computations that distinguish fixed-set failure from existential five-set success.

3 Exact finite ground truth

All exact computations use

$$d(G) = |E(G)| - \text{MaxCut}(G).$$

As a regression computation, all 12,172 triangle-free isomorphism classes on ten vertices were checked exactly, with

$$\max d(G) = 4,$$

attained by the balanced C_5 blow-up B_2 . In view of [2], this computation is not a novelty claim.

4 Balanced and mixed extension certificates

The first reduction studies finite extension models organized around the five-part structure of balanced C_5 blow-ups.

4.1 Balanced-extension certificate

For $t = 1, \dots, 25$, the exact C++ checker `scripts/check_balanced_extension.cpp` enumerates the reduced balanced-extension states. In every batch the reported maximum gap is zero.

Proposition 1 (Balanced-extension certificate). *For every balanced-extension state represented by the reduction at $t = 1, \dots, 25$, the target inequality checked by the exact program holds. No violating state occurs in the enumerated reduced state space.*

4.2 Exact mixed $t = 2$ closure

The mixed $t = 2$ model contains 4,013,521 maximal-pair states, of which 2,768,154 are genuinely mixed. Exact MaxCut evaluation gives maximum deletion distance 9 over the full state space and maximum 8 among the genuinely mixed states. A maximal-completion monotonicity argument reduces arbitrary represented legal states to maximal triangle-free completions.

Proposition 2 (Mixed $t = 2$ certificate). *Within the exact mixed $t = 2$ extension model, all represented legal states are closed by maximal triangle-free completion. Among maximal states the largest deletion distance is 9, and among genuinely mixed states it is 8.*

These certificates motivated allowing the coloring of the core graph to change in a coordinated way.

5 Coordinated flips

Fix a five-set $X \subseteq V(G)$ and put $H = G - X$. Let c be an optimal 2-coloring of H . For an assignment a on X , let $E_c(a)$ denote the number of monochromatic edges contributed by $G[X]$ and $E(X, H)$ before modifying the core coloring.

For $v \in V(H)$ define

$$w_a(v) = \#\{\text{monochromatic edges from } v \text{ to } X\} - \#\{\text{bichromatic edges from } v \text{ to } X\},$$

and

$$\sigma_c(v) = \#\{\text{bichromatic core edges incident with } v\} - \#\{\text{monochromatic core edges incident with } v\}.$$

For $S \subseteq V(H)$, let $m_c(S)$ be the number of monochromatic edges of $H[S]$ under c .

Lemma 1 (Coordinated-flip identity). *If all colors on S are flipped, then*

$$g_c(a, S) = \sum_{v \in S} (w_a(v) - \sigma_c(v)) + 2e(H[S]) - 4m_c(S)$$

satisfies

$$b_G(c^S \cup a) - d(H) = E_c(a) - g_c(a, S),$$

where $b_G(\cdot)$ is the number of monochromatic edges under the displayed coloring.

Proof. Each boundary edge incident with a flipped vertex changes according to the signed contribution $w_a(v)$. Each core edge with exactly one endpoint flipped changes according to $-\sigma_c(v)$ after summing over vertices. Internal edges of $H[S]$ are counted twice by the vertex sum and require the correction $2e(H[S]) - 4m_c(S)$. Collecting these contributions gives the formula. $\square$

The implemented admissible flip family consists of independent sets and sets inducing complete bipartite subgraphs, together with their complements.

Let μ_X be the uniform distribution on optimal 2-colorings of $G[X]$, counting color reversals separately. For fixed optimal c ,

$$\mathbb{E}_{a \sim \mu_X} E_c(a) = d(G[X]) + \frac{e(X, H)}{2}.$$

Define

$$\Phi_X = \max_{c \text{ optimal on } H} \mathbb{E}_{a \sim \mu_X} \max_{S \text{ admissible}} g_c(a, S).$$

Proposition 3 (Coordinated-flip selection bound). *For every five-set X ,*

$$d(G) - d(G - X) \leq \left\lfloor d(G[X]) + \frac{e(X, G - X)}{2} - \Phi_X \right\rfloor.$$

Proof. For each assignment a , choose an admissible flip maximizing the gain. The resulting coloring gives an upper bound on $d(G)$. Average over $a \sim \mu_X$ and then maximize over the optimal core coloring c . Integrality gives the floor. $\square$

5.1 Bichromatic biclique corollary

Write

$$x_a(v) = w_a(v) - \sigma_c(v).$$

If $S = A \cup B$ induces a complete bipartite subgraph of H and every internal edge is bichromatic under c , then $m_c(S) = 0$ and $e(H[S]) = |A||B|$. Hence the flip identity immediately gives

$$g_c(a, S) = \sum_{v \in S} x_a(v) + 2|A||B|.$$

The pair and star formulas are the cases $K_{1,1}$ and $K_{1,t}$. These are exact bookkeeping corollaries rather than separate extremal results, but they are useful for identifying interaction gain that is invisible in singleton flips.

6 The exact empty-flip threshold

Define

$$\ell(X) = 2d(G[X]) + e(X, G - X), \quad L(G) = \min_{|X|=5} \ell(X).$$

Setting $S = \emptyset$ gives $\Phi_X \geq 0$ and therefore

$$d(G) - d(G - X) \leq \left\lfloor \frac{\ell(X)}{2} \right\rfloor.$$

Proposition 4 (Automatic five-set threshold). *For a triangle-free graph on $5k$ vertices, if a five-set X satisfies*

$$\ell(X) \leq 4k - 1,$$

then

$$d(G) - d(G - X) \leq 2k - 1.$$

In particular, Candidate A is automatic whenever $L(G) \leq 4k - 1$.

Remark 1 (Correction to Version 1). *At $k = 3$, the automatic threshold is $L(G) \leq 11$, not $L(G) \leq 10$. Thus global $L(G) = 11$ is not a genuinely hard Candidate-A regime. The $L = 11$ computations below remain valid as experiments on exchange structure, but they are not required to prove the five-set inequality.*

7 Fixed-set failure and exchange diagnostics

Early experiments suggested stronger fixed-five-set gain statements. A direct actual-core analysis produced realizable residual configurations for which these stronger gain thresholds fail. Importantly, failure of a strong fixed- X statement does not imply failure of the existential five-set induction target.

For 22 residual witnesses, an exact audit over all $\binom{15}{5} = 3003$ five-sets gives

$$\max q_{\min} = 4, \quad q(Y) = d(G) - d(G - Y),$$

and restricting to 3-for-3 exchanges relative to the original set gives restricted minima distributed as

$q_{\min}$	1	2	3	4
#	1	10	8	3.

Thus every one of these explicit fixed-set obstructions is repaired by changing the five-set.

8 The saved global- $L = 11$ 3-for-3 audit

A 3-for-3 exchange of a recorded five-set X is a five-set Y satisfying $|X \cap Y| = 2$. On fifteen vertices there are

$$\binom{5}{2} \binom{10}{3} = 1200$$

such exchanges.

The saved corpus contains 3,436 canonical records with global $L(G) = 11$. Relative to the distinguished recorded five-set, every one of these records admits a 3-for-3 exchange Y with

$$d(G) - d(G - Y) \leq 5.$$

Among them, 762 have $d(G) > 5$ and require a nontrivial search; all 762 pass.

Remark 2 (Interpretation after the threshold correction). *Because $L(G) = 11$ already lies in the automatic range at $k = 3$, this audit is not a proof certificate needed for Candidate A. Its value is structural: it demonstrates, on a precisely scoped saved corpus, that obstructions to stronger fixed-set statements can be repaired by a small exchange of the deletion set.*

The scope audit independently matches the 3,436 global- $L = 11$ IDs and canonical graph6 strings against the exchange file. The statement is exhaustive only for this saved global- $L = 11$ corpus; no claim is made here that the parent saved corpus is the full universe of triangle-free graphs on fifteen vertices.

9 Criticality and odd-cycle-cover language

The edge-bipartization number is the minimum size of an edge set meeting every odd cycle. The following two elementary facts are useful when considering a hypothetical edge-minimal counterexample.

Lemma 2 (Critical edge and minimum transversal). *For an edge e ,*

$$d(G - e) = d(G) - 1$$

if and only if e belongs to some minimum odd-cycle edge transversal of G .

Proof. If a minimum transversal T contains e , then $T \setminus \{e\}$ bipartizes $G - e$, so $d(G - e) \leq d(G) - 1$; deleting one edge can reduce d by at most one, hence equality holds. Conversely, if $d(G - e) = d(G) - 1$, a minimum transversal of $G - e$, together with e , is a minimum transversal of G containing e . $\square$

Lemma 3 (Private odd-cycle witness). *Let T be an inclusion-minimal odd-cycle edge transversal. For every $e \in T$ there exists an odd cycle C_e with*

$$C_e \cap T = \{e\}.$$

If G is triangle-free, then $|C_e| \geq 5$.

Proof. If no such cycle existed, then every odd cycle meeting e would also meet $T \setminus \{e\}$, so $T \setminus \{e\}$ would still meet every odd cycle, contradicting inclusion-minimality. $\square$

These observations belong naturally to the existing theory of frustration-critical signed graphs [3]. They are included to connect the five-set induction project with that language, not as novelty claims.

Weakly bipartite signed graphs provide another relevant viewpoint. The odd-circuit-cover polyhedron is integral exactly in the odd- K_5 -minor-free class by Guenin's theorem [4, 5]. Any future LP-based attack on the Erdős conjecture must therefore distinguish genuinely new triangle-free consequences from this established signed-graph theory.

10 Reproducibility

The principal release artifacts are:

- exact balanced- and mixed-extension checkers;
- `results/hard_regime_L_n15/L_records.tsv`;
- `results/hard_regime_L_n15/L11_exchange.jsonl`;
- `scripts/build_L11_3for3_input.py`;
- `scripts/check_L11_3for3_all.cpp`;
- deterministic audit outputs and SHA-256 checksums.

The computational claims are intended to be reproducible from explicit finite inputs. We distinguish throughout between symbolic identities, finite exact certificates, and conjectural structural statements.

11 Discussion

The main lesson of the computation is negative as well as positive. Strong fixed-five-set statements can fail for realizable configurations, while the existential induction target remains true after changing the five-set. This suggests that a successful proof, if it follows this architecture, should prioritize selection over increasingly elaborate certification of a distinguished set.

The corrected floor threshold further simplifies the picture: at $k = 3$, all global- $L = 11$ records are already automatic for Candidate A. The exchange audit should therefore be viewed as a diagnostic experiment on selection geometry rather than as evidence needed to close that regime.

The critical signed-graph and weakly-bipartite viewpoints provide a useful external check on novelty. Edge-criticality, private odd-cycle witnesses, and odd- K_5 integrality obstructions are part of established theory. A future theorem paper should therefore contribute a genuinely new triangle-free structural statement connecting these tools to the sharp $N^2/25$ bound.

Status and non-claims

The Erdős triangle-free bipartization conjecture remains open in general. This manuscript does not claim to solve it. The finite computations at $n = 10$ and $n = 15$ are not claimed as new extremal evaluations. The global- $L = 11$ 3-for-3 audit is a structural computation, not a necessary proof of Candidate A after the corrected threshold. The manuscript's current contribution is a reproducible five-set induction framework, exact flip identities, reduced-state certificates, and explicit diagnostics of false strengthenings.

References

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